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# ESEC/FSE'17

**Proceedings of the 2017 11th Joint Meeting on**

# Foundations of Software Engineering

*Edited by:*
**Eric Bodden, Wilhelm Schäfer, Arie van Deursen, and Andrea Zisman**

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# Message from the Chairs

On behalf of the entire organizing team of ESEC/FSE 2017 it is a great honor and pleasure for us to welcome you to the 11th Joint Meeting of the European Software Engineering Conference and the ACM SIGSOFT Symposium on the Foundations of Software Engineering held September 4th-8th, 2017 in Paderborn, Germany.

The conference brings together researchers, practitioners, and educators in software engineering to discuss the most recent innovations and challenges in the area. This year's program continues the great tradition of previous ESEC/FSE joint meetings in being as exciting as it is diverse.

The research track attracted 295 submissions from all over the world. The reviewing process was performed by a program committee composed of 39 of the world's top software engineering experts from academia and industry. This year, for the first time in the history of ESEC or FSE, we followed a double-blind reviewing process. Each paper received reviews from three members of the programme committee, as well as one short meta-review explaining the eventual decision. Authors of the submitted papers were allowed to provide a response to the reviews during the rebuttal phase, followed by an extensive online discussion of the papers by the reviewers. A physical PC meeting was held in Buenos Aires on May 27-28, right after ICSE 2017. The committee eventually accepted 72 papers covering a large variety of software engineering topics, including empirical studies, program analysis, testing, models, development techniques, adaptive systems, search-based software engineering, security and privacy, among others.

This year's ESEC/FSE again encouraged authors of accepted papers to submit data sets and any additional artefacts needed to replicate their work to the Artifact Evaluation track. This year, the track implements ACM's "Result and Artifact Review and Badging" policy, which recognizes high quality artifacts through official badges. The "Artifacts Evaluated - Functional" badge indicates that an artifact is complete, well-documented and allows to obtain the research track paper's results, while the "Artifacts Evaluated - Reusable" badge indicates that an artifact is of such a high quality that it can be reused as is on other datasets, or for other purposes. Of course, every artifact had to be easy to use or analyze, and should be accessible from a public archival repository for at least 5 years. Any research track paper with an accepted artifact benefits from an extra page in the conference proceedings dedicated to the presentation of the artifact (in addition to being highlighted in the conference program).

Out of the 72 accepted research track papers (each eligible to submit to the artifacts track), 13 artifact submissions were received, ranging from data sets to benchmarks or tools. Each artifact was reviewed by 3 artifact PC members (out of a committee of 17 international experts), after which the authors were able to react to these reviews by fixing their artifacts or their documentation. Subsequent re-reviewing and discussion led to 11 artifacts being accepted, 6 of which obtaining the functional badge, while 5 were of such high quality that they obtained a reusable badge. Given that the whole reviewing procedure (including rebuttal phase) took place in the time span of 3 weeks, we would like to thank the authors and reviewers for their dedication! Finally, a special word of thanks to the authors of the reusable badge artifacts, for their extra effort in advancing the state-of-the-art in the community.

ESEC/FSE 2017 continues to support the Journal First Initiative initiated some years ago in the SIGSOFT community. With this initiative, selected journal papers that are not extensions of conference papers are presented at one of the leading conferences in the area. This allows attendees to hear talks about excellent research work from world-leading journals that has only previously appeared in a journal and would, therefore, otherwise go without being presented at any conference. This year, the conference invited the authors of all eligible papers of the ACM Transactions On Software Engineering and Methodology (TOSEM), the IEEE Transactions on Software Engineering (TSE), and the Springer Journal on Empirical Software Engineering (EMSE) to present. This resulted in the invitation of 35 Journal First papers, from which authors of 25 papers accepted the invitation.

The conference also includes an industry track with three sessions where 17 papers, selected out of 36 submissions, are presented with lively discussions; one tool demo track with 14 demonstrations selected out of 26 submissions; one student research competition with the participation of 8 candidate students; and a doctoral symposium track, to be held before the conference, in which 4 research proposals were discussed.

The conference features altogether four invited keynote talks. Two keynote talks for the research track entitled *The Rising Tide Lifts all Boats: The Advancement of Science in Cyber Security* by Laurie Williams (North Carolina State University), and *Verifying the Forecast: How Climate Models are Developed and Tested* by Steve Easterbrook (University of Toronto); one keynote from the *SIGSOFT Impact Award,* entitled *Reflections on the REST Architectural Style and Principle Design of the Modern Web Architecture* by Roy T. Fielding (Adobe Systems Inc.) and Richard N. Taylor (University of California Irvine); and one keynote talk for the Industry Track entitled *Software Engineering Research Results in Industrial Practice - a Tale of two Projects* by Wolfgang Emmerich (Zühlke U.K.).

Besides the traditional doctoral symposium, the pre-conference activities include four workshops on a set of important topics: the 8th Workshop on Automated Software Testing (A-TEST), the Workshop on Software Engineering and Digital Forensics (SERF), the 3rd International Workshop on Software Analytics (SWAN), and the 2nd International Workshop on App Market Analytics (WAMA). The conference is also co-located with the Annual Symposium on Search-Based Software Engineering (SSBSE), to be held right after ESEC/FSE.

We are extremely grateful for the various people who made this great event possible. Much gratitude and our best wishes go to Wilhelm Schäfer, who as the original general chair was instrumental in bringing ESEC/FSE 2017 to Paderborn and in securing the conference venue and initiating the composition of the organizing committee. Quite unfortunately, in Spring 2016 a sudden illness forced Wilhelm to suspend his engagement in the conference organization. After a brief discussion with the Steering Committee, Eric Bodden agreed to take over the remaining organization as a co-chair. The entire Organizing Committee sends Wilhelm his best wishes on his way to his recovery.

Many other people were instrumental in the organization, and we apologize in advance for not mentioning everyone by name. We are extremely thankful to the members of the Organizing Committee who worked hard to the development of all parts of the conference. We extend a special thanks to the Program Committee members of all tracks. We would like to give a special thanks to the members of the Programme Committee of the research track who worked extremely hard to review the many submissions, participated in the online discussions, and attended the physical PC meeting. We would

like to thank Sebastian Uchitel, the ICSE 2017 General Chair, for his assistance with arrangements for the physical research track PC meeting. We are indebted to the local support staff of the Heinz Nixdorf Institute and Heinz Nixdorf MuseumsForum, who were extremely helpful and professional in all organizational matters.

We would like to acknowledge ACM for the sponsorship of ESEC/FSE 2017 through its Special Interest Group on Software Engineering (SIGSOFT). We also deeply thank our supporters Miele, Google, Oracle, Rohde & Schwarz CyberSecurity and Adesso. With this sponsorship, financial support, and many contributions from members of the software engineering community, we are happy to present to you an exciting and rich program with ideas that we are sure will have great impact and will foster progress in software engineering worldwide.

Finally, we want to thank all of the conference attendees for contributing to making ESEC/FSE 2017 a success. We hope that you will find the program interesting, thought-provoking, and inspiring, and that the conference will give you valuable opportunities to share ideas with researchers and practitioners from the vibrant software engineering community.

*Eric Bodden, General Co-Chair (with Wilhelm Schäfer)*
*Andrea Zisman & Arie van Deursen, Program Co-Chairs*

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